📄 AI-BFSI Internship Project Documentation

Project Demo Planning – Subscription Waste Detector

1. Introduction

A project demonstration is the final stage of the software development lifecycle where the completed application is presented to evaluators to verify that the implemented features satisfy the defined objectives.

For the **Subscription Waste Detector**, the demonstration showcases Version 1 features: expense data upload, recurring subscription detection, interactive dashboard visualization, renewal reminders, and AI-powered recommendations. The demo follows a structured sequence to ensure clarity and systematic presentation.

2. Objectives of the Project Demonstration

- Present the completed application to evaluators
- Demonstrate implemented functional features
- Validate alignment with project objectives
- Showcase AI integration via Groq API
- Explain overall workflow
- Highlight practical value of the solution

3. Demonstration Scope

Included

- Launch Streamlit application
- CSV file upload
- Excel file upload
- Manual expense entry
- Expense data processing
- Subscription detection
- Subscription listing
- Monthly spending analysis
- Plotly visualizations
- Renewal reminders
- AI-powered recommendations

Excluded

(Not part of Version 1)

- Authentication

- Database integration
- Cloud synchronization
- Bank API integration
- Email notifications
- Mobile application
- Multi-user support

4. Demonstration Environment

Component	Description
Application	Subscription Waste Detector
Platform	Streamlit
Programming Language	Python
Browser	Google Chrome
Development Environment	Visual Studio Code
AI Service	Groq API
Sample Data	CSV and Excel expense files

5. Demonstration Preparation

- Verify Python and libraries installed
- Configure Groq API key
- Ensure Streamlit launches successfully
- Prepare sample CSV/Excel datasets
- Confirm internet connectivity for AI recommendations
- Check dashboard components render correctly

6. Demonstration Workflow

```
Start Demonstration
        |
        ▼
Introduce Project
        |
        ▼
Explain Problem Statement
        |
        ▼
Launch Streamlit Application
        |
        ▼
Upload Expense Data
        |
        ▼
Process Transactions
        |
        ▼
Detect Subscriptions
        |
        ▼
Display Dashboard
        |
        ├────────▶ Plotly Visualizations
```

```
│
└────────► Groq API
               │
               ▼
     AI Recommendations
               │
               ▼
      Explain Results
               │
               ▼
         Conclusion
```

7. Feature Demonstration Sequence

Step	Activity	Expected Outcome
1	Launch application	Streamlit interface opens successfully
2	Upload CSV file	Expense records imported
3	Upload Excel file	Spreadsheet processed correctly
4	Manual expense entry	New expense record accepted
5	Data processing	Transactions prepared for analysis
6	Subscription detection	Recurring subscriptions identified
7	Dashboard presentation	Financial summary displayed
8	Plotly visualization	Interactive charts generated
9	Renewal reminder	Upcoming renewals displayed
10	AI recommendations	Groq API generates personalized suggestions

8. Demonstration Script

- **Step 1 – Project Introduction:** Brief overview of the Subscription Waste Detector and its purpose
- **Step 2 – Problem Statement:** Explain recurring subscription challenges
- **Step 3 – Solution Overview:** Introduce Streamlit application workflow
- **Step 4 – Live Demonstration:** Show CSV/Excel upload, manual entry, detection, dashboard, charts, reminders, AI recommendations
- **Step 5 – Explain Results:** Interpret outputs and recommendations
- **Step 6 – Project Summary:** Conclude by linking features to problem resolution

9. Expected Demonstration Outcomes

- Application launches successfully
- Expense data processed correctly
- Recurring subscriptions identified
- Dashboard displays summaries accurately
- Plotly charts render smoothly
- Renewal reminders shown
- AI recommendations generated via Groq API

10. Risks During Demonstration

Risk	Mitigation

Invalid input files	Use verified sample datasets
Missing internet connectivity	Check connection before demo (Groq API requires internet)
Incorrect file format	Use supported CSV/Excel formats
Missing dependencies	Install libraries beforehand
Runtime errors	Test application thoroughly before live demo

11. Required Demonstration Resources

- Computer with Python installed
- Visual Studio Code
- Streamlit application
- Google Chrome browser
- Sample CSV/Excel files
- Internet connection for Groq API

12. Success Evaluation

The demo is successful if:

- All implemented features operate correctly
- Subscription analysis is accurate
- Visualizations display properly
- AI recommendations retrieved successfully
- Workflow completes without errors

13. Conclusion

The **Project Demo Planning** document provides a structured approach for presenting the Subscription Waste Detector. By following a logical sequence—from introduction and problem explanation to live feature demonstration and result interpretation—the demo effectively showcases Version 1 functionality.